Hill Cipher

requirments

- fixed block size of input
- dynamic generating the key matrix by the user

Solution:

plaintext: LOG , there will be 3x3 matrix

Key matrix:

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

follow the letters into numbers using

A=0, B=1, ..., Z=25

$$LOG = \begin{bmatrix} 11 \\ 14 \\ 6 \end{bmatrix}$$

+ Encryption

formula: $C = KP \bmod 26$

$$\Rightarrow C = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 11 \\ 14 \\ 6 \end{bmatrix} \bmod 26$$

calculate each row

$C_1 = 6(11) + 24(14) + 1(6) = 408$

$408 \bmod 26 = 18$

$C_2 = 13(11) + 16(14) + 10(6) = 427$

$427 \bmod 26 = 11$

$C_3 = 20(11) + 17(14) + 15(6) = 548$

$548 \bmod 26 = 2$

Therefore

$$C = \begin{bmatrix} 18 \\ 11 \\ 2 \end{bmatrix}$$

convert numbers to letters:

18 = S

11 = L

2 = C

Therefore ciphertext = SLC

+ Decryption

ciphertext = SLC

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$, formula : $P = K^{-1}C \mod 26$

step 1 : Convert ciphertext to numbers

A=0, B=1, ..., Z= 25

$$C = \begin{bmatrix} 18 \\ 11 \\ 2 \end{bmatrix}$$

Step 2 : Find the inverse key matrix

$$K^{-1} = \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix}$$

Step 3 : Apply the decryption formula

$$\Rightarrow P = \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \begin{bmatrix} 18 \\ 11 \\ 2 \end{bmatrix} \mod 26$$

calculate each row :

$P_1 = 8(18) + 5(11) + 10(2) = 219$

$219 \mod 26 = 11$

$P_2 = 21(18) + 8(11) + 21(2) = 508$

$508 \mod 26 = 14$

$P_3 = 21(18) + 12(11) + 8(2) = 526$

$526 \mod 26 = 6$

Therefore

$$P = \begin{bmatrix} 11 \\ 14 \\ 6 \end{bmatrix}$$

step 4 : Convert numbers back to letters

11 = L

14 = O

6 = G

Therefore Plaintext = LOG